Mechanistic Validity: A Theory of Validity for Interpretability Claims

Abstract

A mechanistic explanation of a neural network is not a single experimental result, but a structured claim linking a theoretical construct, a measurement procedure, an intervention, and a scope of interpretation. We introduce Mechanistic Validity (MechVal), a formalized epistemology for validating these causal claims, drawing on validation methodology from causal inference, measurement theory, and the philosophy of science. The framework makes interpretability claims explicit, falsifiable, and comparable, formalizing the discovery, validation, and interpretation of causal mechanisms as separate evidential stages. MechVal evaluates claims through six layers: (1) *description mode*, (2) *evidence family*, (3) *metrics*, (4) *criteria*, (5) *validity type*, (6) *verdict*. A gap at any layer is visible in the final verdict. The framework is method-agnostic, applying the same criteria to circuits, SAE features, probing classifiers, steering vectors, transcoders, and crosscoders. Applying the framework to 13 published circuits, we show that the tier system discriminates among claims—from Proposed (probing classifiers) to Triangulated (induction heads)—with every verdict traceable to specific present or missing evidence. By making the evidential status of a claim explicit and the path to strengthening it concrete, MechVal provides a foundation for interpretability claims that are explicit, testable, and scientifically grounded.

1 Introduction

Mechanistic interpretability has produced a remarkable catalog of internal mechanisms: induction heads that implement in-context copying (Olsson et al., 2022), a multi-role circuit for indirect object identification (Wang et al., 2023), a greater-than algorithm (Hanna et al., 2023), and sparse autoencoder features proposed to correspond to human-interpretable concepts (Cunningham et al., 2023). The field has also begun to build evaluation infrastructure: faithfulness metrics for circuits (Miller et al., 2024), the Mechanistic Interpretability Benchmark for comparing localization methods across tasks and models (Mueller et al., 2025), SAEBench for auditing sparse autoencoder quality (Karvonen et al., 2025), and causal scrubbing for testing mechanistic hypotheses via behavior-preserving resampling (Chan et al., 2022).

These contributions evaluate important objects—methods, circuits, features, alignments, benchmark scores. They do not by themselves answer a different question: what has been established by a mechanistic interpretability *claim*? A claim such as "this head is a name mover," "this feature represents deception," or "this subspace implements a causal variable" is not just a circuit or a metric value. It links a theoretical construct to a measurement procedure, a causal intervention, a scope of generalization, and a natural-language interpretation. Shared metrics are not yet shared validity criteria.

A mechanistic narrative that was retrofitted to match ablation results is currently indistinguishable from one that has survived refutation attempts. This matters because a mechanistic explanation has many adjustable parts—which components to include, what roles to assign, which edges to draw—and a narrative with enough flexibility will always fit the observations. The MechVal framework introduced below provides the criteria needed to distinguish well-supported claims from plausible-sounding ones, making the gaps in evidence visible rather than leaving them implicit.

Several recent results sharpen this point. Sutter et al. (2025) prove that without structural constraints on alignment maps, causal abstraction becomes vacuous: sufficiently expressive maps can perfectly align randomly initialized language models to arbitrary algorithms, achieving 100% interchange-intervention accuracy on tasks the models cannot solve. The SAEBench audit reveals that two of six standard SAE evaluation metrics fail basic diagnostics for reseed stability, ground-truth correlation, and discriminability (Karvonen et al., 2025). And execution-grounded reproducibility checks via MechEvalAgent (Bai et al., 2026) surface claim-evidence coherence issues that narrative review misses. Each finding bears on a different link in the inference chain: alignment-map complexity, measurement reliability, and reproducibility, respectively. What is missing is not any one fix but a principled way to ask, of any claim, which links have been established and which have not.

We introduce MechVal, a formalized epistemology for validating mechanistic interpretability claims, drawing on validation methodology from causal inference, measurement theory, and the philosophy of science. The framework makes interpretability claims explicit, falsifiable, and comparable, formalizing the discovery, validation, and interpretation of causal mechanisms as separate evidential stages. It defines 27 operational criteria across five validity types—construct, measurement, internal, external, and interpretive—in dependency order, with five verdict tiers marking qualitative transitions in evidential status.

MechVal is claim-centric and method-agnostic: it evaluates circuits, SAE features, probing classifiers, steering vectors, transcoders, and crosscoders with the same criteria. A faithfulness score, an interchange-intervention accuracy, a probe accuracy, or a causal-scrubbing result can all serve as evidence within the framework; the framework specifies what kind of evidence each provides, what kind of claim it can support, and what remains untested. MechVal does not replace existing benchmarks or methods. It provides a common layer above them: claim-level validation built on method-level evaluation.

We describe the framework (Section 4), its theoretical foundations (Section 5), its application to 13 published circuits (Section 6), and its implications for the field (Section 7).

2 Related Work

Three recent efforts assess different layers of the MI evidence stack, and MechVal is designed to complement rather than replace them.

The Mechanistic Interpretability Benchmark (MIB; Mueller et al., 2025) compares circuit-discovery and causal-variable-localization methods on a shared task suite using Circuit Model Distance and Interchange Intervention Accuracy. MIB is method-vs-method: given a task, which discovery algorithm produces the best circuit? MechVal is method-agnostic and claim-centric: given a claim about how a model works, what evidence would be needed to sustain it? The two are stacked rather than competing—MIB scores a method's output on a single dimension, while MechVal assesses how that output is interpreted, generalized, and reported.

SAEBench (Karvonen et al., 2025) evaluates sparse autoencoders along six metric families and audits the metrics themselves. Two of six metrics (TPP, SCR) fail diagnostic tests for reseed stability, ground-truth correlation, and discriminability, motivating the explicit measurement-validity dimension in our framework. MechVal treats SAEBench's audit as evidence *about* the M-dimension criteria (M1 reliability, M5 sensitivity, M6 invariance) rather than as a competing benchmark.

MechEvalAgent (Bai et al., 2026) performs execution-grounded reproducibility checks on published MI papers, finding that 93% fail to reproduce when their code is run and that 80% fail claim-evidence coherence checks. Where MechEvalAgent asks "does this code run and match its prose?," MechVal asks "does this evidence pattern support this kind of claim?"

Finally, the LLM-as-judge paradigm (Zheng et al., 2023) has been applied to open-ended evaluation of MI research. However, unstructured LLM evaluation lacks the operational criteria needed to distinguish genuine validity from surface plausibility—the same limitation that affects human narrative review, as demonstrated by MechEvalAgent's finding that narrative review misses issues that execution-grounded checks catch.

A companion paper automates MechVal via structured LLM extraction against the 27 criteria, and describes protocols (curated experiment designs that bundle metrics around specific validity questions) and synthesis algorithms (aggregation across evidence families to produce quantitative scores). Automation and scoring are treated separately because the framework's correctness is independent of whether verdicts are assigned by hand or by machine.

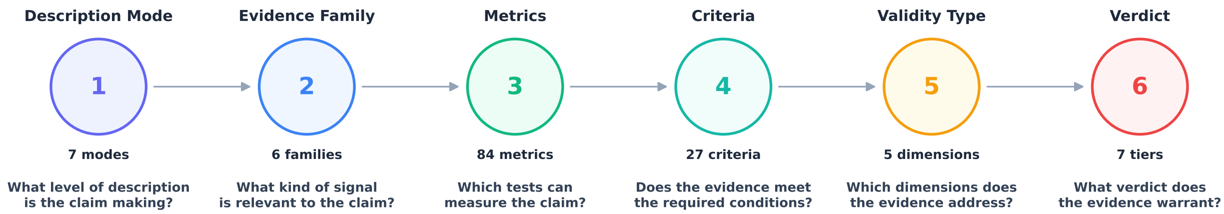

Figure 1: The MechVal framework hierarchy. When running experiments, evidence builds upward from metrics through criteria to a verdict. When auditing a claim, the process reverses: starting from a verdict, the framework identifies which criteria are met and whether the description mode is supported by the available evidence.

3 From Claim to Verdict

Consider a concrete claim: "The IOI circuit uses three name-mover heads (9.9, 9.6, 10.0) to copy the indirect object to the output position." This claim appears in a widely-cited paper, is supported by ablation experiments, and has a clear mechanistic narrative. What has this claim established, and what remains untested?

The framework evaluates such a claim through six layers (Figure 1):

1. **Description mode.** At what level is the claim made? The IOI claim operates at the implementational-functional level: it names specific heads and says what each one does. A weaker claim ("GPT-2 solves IOI") would be computational; a stronger one ("head 9.9 implements copying via its OV circuit") would require additional structural evidence.
2. **Evidence family.** What kinds of evidence are available? The original IOI paper provides primarily causal evidence (ablation) and some behavioral evidence (faithfulness). Structural evidence (weight-space analysis of the OV circuits) and cross-model evidence (does the same circuit appear in GPT-2 Medium?) would strengthen the claim by adding independent failure modes.
3. **Metrics.** Which concrete tests were run? Activation patching, role ablation, faithfulness scores. Each metric belongs to an evidence family and produces a measurement that feeds into criteria evaluation.
4. **Criteria.** Does the evidence meet the bar? The IOI circuit passes necessity (I1: ablation degrades performance) and sufficiency (I2: the circuit alone recovers 87% of logit difference). But specificity (I3: does the circuit affect *only* IOI?) is untested, and the 87% faithfulness number is method-conditional—it drops below 50% under non-mean-ablation methods (Miller et al., 2024), partially failing intervention reach (E1).
5. **Validity type.** Which validity dimensions are satisfied? Construct validity (the claim is well-defined), partial internal validity (necessity and sufficiency shown but specificity untested), partial external validity (limited prompt generalization, no cross-model replication). Measurement validity is unaddressed—no calibration of the ablation metrics themselves.
6. **Verdict.** What tier does the claim reach? Mechanistically Supported—necessity and sufficiency are established with at least one method, but the evidence comes primarily from one methodological family (ablation-based), key criteria remain untested (I3, I4, E4), and the faithfulness result is method-conditional. The verdict is not a judgment on the paper's quality; it is a precise characterization of what has been established and what has not.

The 27 criteria also formalize common failure modes observed in published MI research (Table 1).

Table 1: Twelve common failure modes in MI research, each mapped to the criteria that detect it.

Failure mode	**What happens**	**Criterion**	**Example**
Cherry-picking metrics	Run multiple ablation methods, report only the best number	E1 Intervention reach	IOI: 87% mean ablation reported; resample ablation (40%) omitted
No baseline control	Report high score without testing a random or untrained model	M2 Baseline separation	Nonlinear alignment maps achieve 95% IIA on random models (Sutter et al., 2025)
No negative controls	Only test predictions that should pass; never test what the mechanism should *not* affect	I3 Specificity C1 Falsifiability	IOI: no test of whether the circuit also degrades SVA or factual recall
Construct incoherence	Target construct is not separable from a neighboring one	C4 Discriminant validity	Gender bias circuits: bias and knowledge share the same heads
Off-target effects	Ablation works for the target task; other tasks never checked	I3 Specificity	Knowledge neurons: editing one fact corrupts related facts (ripple effects)
Circuit redefinition	Drop heads that hurt faithfulness, call it “refined”	C1 Falsifiability	Adjusting circuit membership to maximize the target metric
Post-hoc role labeling	Name heads after observed behavior; the label cannot fail	C1 Falsifiability	“Name mover” assigned after seeing logit attribution
No double dissociation	Show necessity but never test the converse	I4 Double dissociation	IOI circuit ablated $\rightarrow$ IOI breaks; SVA circuit $\rightarrow$ IOI intact never tested
Interpretive inflation	Label implies intent or algorithm beyond what evidence shows	V4 Anthropomorphism check	“World model” for Othello (evidence supports “linearly decodable”)
Scope creep	Validate on one prompt format, claim generality	V5 Scope decl. E2 Prompt gen.	“The IOI circuit” tested only on two-name templates
Flexible scope	Narrow the claim after seeing results so it cannot fail	V5 Scope declaration	Restricting scope post-hoc to the subset of prompts that work
Ignoring alternatives	Circuit works, but so does a different set of heads	I4 Double dissoc. C3 Convergent val.	Meloux et al. find alternative IOI circuits with comparable faithfulness

4 The Framework

The MechVal framework has six layers, presented here in pipeline order (Figure 1): description modes (Section 4.1), evidence families (Section 4.2), metrics (Section 4.3), criteria (Section 4.4), validity types (Section 4.5), and verdict tiers (Section 4.6).

4.1 Description Modes

A description mode is a commitment about what kind of explanation is being offered (Table 2). It is not a label applied after analysis—it is a declaration that determines what counts as evidence, what counts as a gap, and what the finding means if it succeeds. The distinction matters because mechanistic interpretability routinely conflates levels: a paper that discovers *which components* are active (implementational-topographic) writes a narrative about *what the model computes* (computational). Each level upgrade requires additional bridging evidence that the lower level cannot provide.

The seven modes, adapted from Marr's hierarchy (Marr, 1982), form a partial order by evidential commitment. A computational claim commits to a function being computed and a reason it is the right function. An algorithmic claim commits to a specific procedure. A representational claim commits to what information is encoded but not how it is processed. The four implementational sub-types commit to progressively weaker structural facts: which components are involved (topographic), how they are wired (connectomic), what their activations look like (statistical), and what input-output transformation each performs (functional).

Table 2: Seven description modes, ordered from most committal (computational) to least committal (statistical). Higher modes require all evidence from lower modes plus bridging evidence.

	Mode	What it asks	Example
1	Computational	What function does the system compute, and why?	"GPT-2 predicts the indirect object"
2	Algorithmic	What procedure does it execute?	"A detect-inhibit-copy algorithm"
3	Representational	What information is encoded, and in what geometry?	"The IO name is linearly encoded at layer 8"
4	Impl.-Functional	What transformation does each component perform?	"Head 9.9 copies names to the output"
5	Impl.-Connectomic	How are components wired?	"S-inhibition feeds name movers via residual stream"
6	Impl.-Statistical	What do activations look like?	"Head 9.9 attends strongly to IO position"
7	Impl.-Topographic	Which components are involved?	"Heads 9.9, 9.6, 10.0"

The mode assigned to a verdict is the strongest level for which all required evidence is present. Partial evidence at a higher level does not upgrade the mode—it is reported as suggestive while the verdict is issued at the established mode. For example, a claim at the algorithmic level ("the model runs a detect-inhibit-copy algorithm") requires both the topographic evidence (which heads are involved) and the additional evidence that the heads implement the claimed procedure—path-level causal tracing, timing consistency, and sufficiency of the minimal circuit.

4.2 Evidence Families

An evidence family classifies a metric's output by the kind of signal it produces, independent of the specific tool used to generate it (Table 3). The classification matters because convergent validity—two independent lines of evidence agreeing—is the strongest form of support for a mechanistic claim. But not all agreement is equally informative: two causal metrics agreeing shares failure modes, while a causal ablation and a representational probe agreeing is substantially stronger, because convergent evidence provides triangulated support across independent methodologies.

Table 3: Six evidence families. Convergent evidence across families with independent failure modes is qualitatively stronger than redundant evidence from the same family.

	Family	**What it measures**	**Examples**
A	Causal	Effects of interventions on computation	Activation patching, DAS-IIA, CATE, mediation
B	Structural	Weight-space properties (no forward pass)	SVD, effective rank, OV/QK composition
C	Information	Information flow and dependencies	Mutual information, PID, OCSE, transfer entropy
D	Behavioral	Input-output behavior under manipulation	Faithfulness, logit diff, KL divergence, dose-response
E	Representational	Geometry of internal representations	CKA, RSA, linear probes, attention entropy
F	Measurement	Trustworthiness of other metrics	Bootstrap stability, seed variance, convergent validity

The primary function of the family classification is to structure convergent validity assessments. When evaluating a claim, the analyst asks three questions: How many evidence families support the claim? Which families are represented? And are there families that *should* support the claim but do not? Absence of expected evidence from a family is informative: it may indicate a gap in the claim or a limitation of the proposed mechanism.

Each family answers a different question. Causal metrics ask: does this component matter for this behavior? Structural metrics ask: what does the architecture encode before any data flows? Behavioral metrics ask: does the proposed circuit actually produce the behavior it is supposed to explain? Representational metrics ask: what does this component represent? Information-theoretic metrics ask: how much does this component know about the task variable? And measurement metrics ask: can the other metrics' outputs be trusted?

Each family has characteristic strengths and blind spots. Causal metrics (A) establish necessity and sufficiency directly but are vulnerable to backup circuits that compensate after ablation. Structural metrics (B) cannot be confounded by runtime compensation but cannot distinguish used capacity from unused capacity. Behavioral metrics (D) test whether the circuit reproduces the model's outputs, but behavioral equivalence does not imply mechanistic equivalence—different circuits can produce the same behavior for different reasons. Measurement metrics (F) are uniquely important because they evaluate the other families' outputs: a reliable metric pointed at the wrong target produces confident wrong answers.

4.3 Metrics

Metrics are concrete, runnable tests that produce measurements about a neural network—the layer of the hierarchy where empirical contact with the model actually happens. Everything above (evidence families, criteria, validity types, verdicts) depends on what metrics measure and how well they measure it. A metric alone cannot establish a claim; a claim requires evidence from multiple metrics, evaluated against the criteria appropriate to the validity type being asserted.

Each metric takes a model, a task, and optionally an intervention as input, and produces a number as output. Metrics can be composed: one metric's output feeds into another. The 14 calibrations in family F are meta-metrics—they take another metric's output as input and assess whether it is stable (bootstrap stability), reproducible (seed variance), or distinguishable from baselines (baseline separation). A high score on a primary metric is not evidence if the corresponding calibration fails. The current implementation includes 84 primary metrics across families A–E; Table 4 illustrates the causal family. The full inventory for all six families is in Appendix A.3.

Table 4: Causal metrics (family A): effects of interventions on model computation.

Metric	**Description**
A01 SCM / Pearl	Structural causal models and do-calculus as the formal language for circuit claims
A02 Counterfactual DAS	Distributed Alignment Search and IIA for testing causal abstraction hypotheses
A03 Rubin CATE	Conditional average treatment effects across input subpopulations
A04 Woodward	Interventionist theory of causation applied to circuit ablation
A05 MDC / Glennan	New mechanical philosophy: components organized to produce phenomena
A06 Mediation	Natural direct and indirect effects decomposition for circuit paths
A07 Granger / TE	Observational causal discovery via conditional MI and temporal precedence
A08 PID	Decomposing MI into redundant, unique, and synergistic contributions
A09 MDL / SLT	Minimum description length and singular learning theory for circuit complexity
A10 Regularity / INUS	INUS conditions: insufficient but necessary parts of sufficient sets
A11 Actual Cause	Halpern-Pearl actual causation on specific inputs, not just potential causes
A12 Transportability	Formal conditions for circuit generalization across models and distributions
A13 Causal Discovery	NOTEARS / PC algorithms for learning DAGs over circuit components

The causal family draws on multiple distinct theories of causation, each formalized as a metric. Pearl's structural causal models (A01) provide the do-calculus for reasoning about interventions. Geiger et al.'s distributed alignment search (A02) tests causal abstraction hypotheses via interchange interventions. Rubin's potential outcomes framework (A03) estimates heterogeneous treatment effects across input subpopulations. Woodward's interventionism (A04) defines what makes an ablation "surgical" versus confounded. Glennan's new mechanical philosophy (A05) asks whether components are organized to produce the phenomenon. Pearl's mediation analysis (A06) decomposes effects into direct and indirect paths. Granger causality and transfer entropy (A07) measure directed information flow between components across layers. Partial information decomposition (A08) separates redundant, unique, and synergistic contributions. Singular learning theory (A09) measures circuit complexity via the local learning coefficient. Mackie's INUS conditions (A10) identify components that are insufficient but necessary parts of sufficient sets. Halpern-Pearl actual causation (A11) identifies which components were the actual causes on specific inputs, not just potential causes in general. Pearl and Bareinboim's transportability (A12) formalizes when causal findings generalize across settings. And constraint-based algorithms like PC and NOTEARS (A13) learn circuit structure from observational data.

Many metrics correspond to entire experimental methodologies in their source fields; the framework represents them as metrics to provide a common unit of comparison across evidence families and allow them to be scored against the same criteria. The full inventory for all six families is in Appendix A.3.

4.4 Criteria

Criteria are specific, falsifiable conditions that must be met for a validity type to be satisfied (Table 5). Each criterion has a concrete pass condition—for example, I1 (necessity) requires that ablating the component degrades performance across at least two independent methods, not just one. A claim either passes, partially passes, or fails each criterion, producing a structured validity profile rather than a single score.

Table 5: Twenty-seven criteria across five validity types. Each criterion has a concrete test; a claim either passes, partially passes, or fails, producing a structured validity profile.

ID	Criterion	Description
C1	Falsifiability	Can the claim be refuted?
C2	Structural plausibility	Is the mechanism physically possible in the architecture?
C3	Convergent validity	Do multiple independent methods agree?
C4	Discriminant validity	Does the measure distinguish this from neighboring constructs?
C5	Nomological validity	Does the claim fit into a broader theory?
M1	Reliability	Do repeated measurements give the same answer?
M2	Baseline separation	Is the score distinguishable from random/untrained baselines?
M3	Stability	Is the classification robust to perturbation?
M4	Calibration	Are the numbers meaningful?
M5	Sensitivity	Can the instrument detect known-true effects?
M6	Invariance	Does the metric behave consistently across conditions?
I1	Necessity	Is the circuit required for the behavior?
I2	Sufficiency	Is the circuit enough to produce the behavior?
I3	Specificity	Does the circuit do this task and not everything?
I4	Double dissociation	Can this circuit be separated from other circuits?
I5	Confound control	Are alternative explanations ruled out?
E1	Intervention reach	Do different intervention methods agree?
E2	Prompt generalization	Does it work on diverse prompts?
E3	Cross-task generalization	Does the mechanism transfer to related tasks?
E4	Cross-model generalization	Does the mechanism appear in other models?
E5	Graded response	Does partial ablation produce partial effects?
E6	Novel prediction	Does the mechanism predict new, untested behaviors?
V1	Level declaration	At what description mode is the claim made?
V2	Level-evidence match	Does the evidence support claims at that level?
V3	Alternative level	Could the evidence be explained at a different level?
V4	Anthropomorphism check	Is the interpretation projecting human concepts?
V5	Scope declaration	What does the claim explicitly not cover?

The criteria are grouped by validity type but are evaluated independently. A validity type is satisfied only when all its criteria are met; partial satisfaction is reported explicitly. This means a claim can have strong internal validity (I1–I5 satisfied) while failing on measurement validity (M1–M6 unsatisfied)—a common pattern in published MI research, where ablation results are reported without any calibration of the ablation metrics themselves. The structured profile makes these asymmetries visible: instead of a single verdict, the researcher sees exactly which dimensions are strong and which are weak.

Every published claim should report, for each satisfied criterion, which metric satisfied it and what value was obtained. This minimum-reporting rule ensures that verdicts are traceable to specific evidence rather than to aggregate impressions.

4.5 Validity Types

The five validity types form a dependency chain: later types cannot be established without first satisfying earlier ones. The ordering reflects logical precedence: a construct that is not well-defined cannot be reliably measured, an unreliable measurement cannot support a causal claim, a causal claim in one setting cannot be generalized, and an interpretation cannot be evaluated until the underlying evidence is characterized.

$$\text{Construct} \rightarrow \text{Measurement} \rightarrow \text{Internal} \rightarrow \text{External} \rightarrow \text{Interpretive} \tag{1}$$

Each type addresses a distinct question about the claim:

Construct validity. Is the thing being measured well-defined? Before any measurement is taken, the construct must be specified precisely enough that the claim is falsifiable, structurally plausible, and distinguishable from neighboring constructs. A "deception feature" that cannot be distinguished from an "uncertainty feature" fails construct validity regardless of how carefully it is measured. This draws on philosophy of science (Popper's falsifiability, Hempel's operationalism) and psychometrics (construct validity, convergent and discriminant validity).

Measurement validity. Are the instruments trustworthy? Once the construct is defined, the measurement tools used to detect it must be reliable (stable across repetitions), calibrated (separable from random baselines), and invariant (consistent across conditions). A metric that gives different answers when run with different random seeds is not evidence. This draws on psychometrics (test-retest reliability, measurement invariance) and pharmacology (assay validation).

Internal validity. Does the evidence support the causal claim? Given a well-defined construct and trustworthy instruments, the evidence must establish that the identified component is causally involved in the behavior—not merely correlated with it. This requires demonstrating necessity (the component is required), sufficiency (the component is enough), and specificity (the component does this task and not everything). This draws on neuroscience (lesion studies, optogenetics, double dissociation) and causal inference (Pearl, 2009).

External validity. Does the mechanism generalize? Causal evidence on one prompt set, with one ablation method, in one model, establishes internal validity at best. External validity requires that the mechanism generalizes across prompts, ablation methods, related tasks, and ideally across models. This draws on pharmacology (Phase III generalization, dose-response) and causal inference (transportability; Pearl and Bareinboim, 2011).

Interpretive validity. Is the interpretation of the mechanism correct? Finally, the natural-language description of the mechanism must be justified by the evidence. A head called a "name mover" based on direct logit attribution carries theoretical implications beyond what the evidence strictly supports. Interpretive validity requires declaring the description level, matching evidence to that level, and checking for anthropomorphic inflation. This is specific to MI but draws on Marr's levels of analysis (Marr, 1982).

4.6 Verdict Tiers

Claims are assigned to one of five verdict tiers representing qualitative transitions in evidential status (Table 6). Two additional labels—Underdetermined and Disconfirmed—handle special cases. The tiers form a strict hierarchy: each requires all evidence from the tier below plus additional evidence.

Table 6: Verdict tiers with requirements. Each tier subsumes the requirements of all lower tiers.

Tier	**Meaning**	**Requirements beyond previous tier**
Proposed	Structural or representational evidence only	Construct validity criteria met (C1–C5)
Causally Suggestive	Necessity shown, sufficiency not established	Internal validity I1 (necessity via ablation)
Mechanistically Supported	Necessity + sufficiency with consistent methods	I1 + I2 (sufficiency), intervention reach (E1), at least 2 ablation variants
Triangulated	Multiple converging lines of independent evidence	Full internal + external + construct criteria from ≥ 3 evidence families
Validated	All five validity types addressed	Measurement + interpretive validity with explicit baselines
Underdetermined	Evidence consistent with multiple mechanisms	Cannot resolve between rival specifications
Disconfirmed	Fails decisively on a key criterion	Negative result

Every tier carries a specific epistemic meaning: a claim at *Proposed* is not a failed claim but one whose evidential status—and the path to strengthening it—is precisely characterized. The verdict tells researchers exactly what additional evidence would advance the claim: a Proposed claim needs causal intervention (I1) to reach Causally Suggestive; a Causally Suggestive claim needs sufficiency evidence (I2) and method consistency (E1) to reach Mechanistically Supported.

Underdetermined and Disconfirmed serve distinct purposes. Underdetermined means the evidence is consistent with multiple competing mechanisms and the framework cannot resolve between them—this is not a lack of evidence but an excess of explanations, requiring additional experiments that discriminate between rivals. Disconfirmed means a criterion has been decisively failed—for example, when the construct itself turns out to be incoherent (C4), as in the gender bias circuits case study.

5 Theoretical Foundations

The criteria in the evaluation pipeline do not come from nowhere. Each one is grounded in a specific intellectual tradition that developed it for a reason—usually because a field got burned by ignoring it. Falsifiability is a criterion because philosophy of science spent a century learning what happens when claims cannot be disconfirmed. Necessity-plus-sufficiency is a criterion because neuroscience spent decades discovering that necessity alone does not establish implementation. Dose-response is a criterion because pharmacology learned that single-dose measurements hide compensatory dynamics.

We call these traditions *lenses* (Figure 2). A lens is not a method or a metric—it is an analytical vocabulary that shapes which questions you ask and how you interpret the answers. Five lenses map to the five validity types; a sixth—mechanistic interpretability itself—grounds interpretive validity, because no other field faces the specific problem of assigning human-interpretable labels to neural network components.

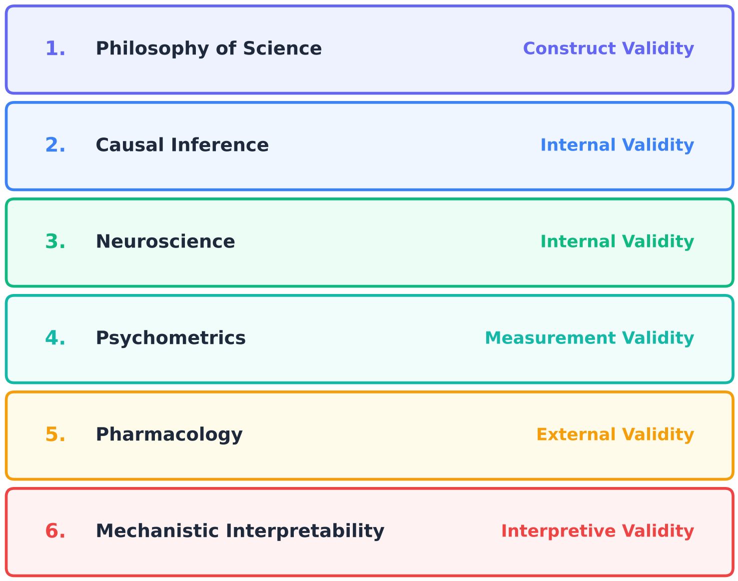

Figure 2: Six theoretical foundations, each grounding a specific validity type. Causal inference and neuroscience both contribute to internal validity through complementary approaches (formal frameworks and experimental designs, respectively).

Each lens answers a core question about the claim: philosophy of science asks *is the entity you named a real thing?* Causal inference and neuroscience ask *does the component implement the computation?* Psychometrics asks *can you trust the number?* Pharmacology asks *does it generalize?* And mechanistic interpretability asks *does the interpretation match the evidence?*

The lens pages are reference material—they explain *why* each criterion exists, not how to apply it. A researcher evaluating a claim follows the pipeline; a researcher who wants to understand why a criterion matters, or who wants to design a new experiment to address a gap, reads the relevant lens.

5.1 Philosophy of Science

The construct validity criteria (C1–C5) draw on Popper's falsifiability criterion, Mayo's severe testing (Mayo, 2018), and Glennan's new mechanical philosophy (Glennan, 2017). A key distinction is between *confirmation* and *corroboration*: a circuit discovered by activation patching and evaluated by activation patching has only been shown to pass the test it was built to pass. Corroboration requires a genuinely risky test the circuit was not optimized for—a behavioral prediction on held-out prompts, a weight-space analysis, a different causal method. The verdict tier system reflects this hierarchy: single-method evidence (Causally Suggestive) is qualitatively weaker than convergent evidence from independent methods (Triangulated).

C5 (nomological validity) requires that the claim fit into a broader theoretical network, following Cronbach and Meehl's nomological network concept. The Duhem-Quine underdetermination problem is directly relevant: multiple circuits with low overlap can each individually pass faithfulness tests, and reporting this explicitly is a stronger finding than suppressing it.

Table 7: Philosophy of science foundations.

Concept	**Source**	**Criterion**	**MI analog**
Falsifiability	Popper 1959	C1	What evidence would refute "this is a name mover"?
Nomological network	Cronbach & Meehl 1955	C5	Does the claim connect to in-context learning theory?
MTMM matrix	Campbell & Fiske 1959	C3, C4	Convergent: probing + patching agree. Discriminant: feature $\neq$ neighboring feature
Underdetermination	Duhem 1906	I4	Multiple circuits explain the same behavior
Severe testing	Mayo 2018	C1	A test that could easily have failed but did not

5.2 Neuroscience

The internal validity criteria (I1–I5) are adapted from neuroscience's toolkit for establishing that a neural circuit implements a computation. I1 (necessity) corresponds to lesion studies; I2 (sufficiency) to stimulation; I4 (double dissociation) is the gold standard for functional specificity (Woodward, 2003). The critical distinction between necessity and sufficiency—well-established in neuroscience but often elided in MI—is central to the framework and marks a qualitative transition between verdict tiers.

A less appreciated requirement is Craver's *mutual manipulability*: constitutive relevance requires not only that intervening on the component changes the behavior, but that intervening on the behavior changes the component's activity. The second direction—does fine-tuning away IOI capability cause name-mover heads to lose their OV copying structure?—is almost never tested in MI, yet it is what "implementation" actually requires. Additionally, the *parcellation problem* is directly relevant: neuroscience learned that the right unit of analysis is defined by convergence across independent modalities, not by any single method. When EAP, weight classification, and ablation disagree on component membership, the "circuit" is probably method-dependent.

Table 8: Neuroscience foundations.

Concept	**Source**	**Criterion**	**MI analog**
Lesion studies	Shallice 1988	I1	Ablation degrades behavior
Double dissociation	Shallice 1988	I4	Circuit A breaks task X not Y; circuit B breaks Y not X
Mutual manipulability	Craver 2007	I3	Fine-tuning away the task changes the component
Multimodal parcellation	Glasser et al. 2016	C3	Circuit boundaries should converge across methods
Neural manifolds	Gallego et al. 2017	V2	Representation geometry constrains algorithmic claims

5.3 Psychometrics

The measurement validity criteria (M1–M6) draw on psychometric test construction. Classical test theory decomposes an observed score into true score plus error ($X = T + E$); M1 (reliability) is the proportion of true-score variance. Generalizability theory extends this by decomposing error into distinct sources—prompt variation, random seed, checkpoint selection—each of which can be quantified and reduced independently.

A crucial insight is that a reliable metric pointed at the wrong target produces confident wrong answers. Sutter et al. (2025) demonstrated this formally: unconstrained nonlinear IIA achieves near-perfect scores on randomly initialized models. The metric is working correctly; it is measuring the alignment map's flexibility, not the model's representational geometry. The framework addresses this through M2 (baseline separation): not "IIA = 0.48" but "$\Delta_{\text{random}} = 0.10$, $\Delta_{\text{untrained}} = 0.15$." A modest absolute number with a large delta is a genuine finding; a large absolute number with a tiny delta is not.

Table 9: Psychometrics foundations.

Concept	**Source**	**Criterion**	**MI analog**
Classical test theory	Lord & Novick 1968	M1	Same metric, same model → same answer
Generalizability theory	Cronbach et al. 1972	M6	Decompose variance into prompt/seed/checkpoint
Signal detection (d')	Green & Swets 1966	M2	Is the score distinguishable from random?
MTMM matrix	Campbell & Fiske 1959	C3, C4	Convergent + discriminant validity
Selectivity	Hewitt & Liang 2019	I3	Probe accuracy minus control accuracy

5.4 Pharmacology

The external validity criteria draw on pharmacology's dose-response methodology. A causally relevant component should produce graded effects: partial ablation should produce partial behavioral change, and the relationship should be monotonic or at least systematic (E5). The framework treats dose-response as a minimum bar for causal claims about magnitude—a requirement absent from most MI evaluations, which typically report binary ablation (all-or-nothing).

The *affinity vs. efficacy* distinction maps with surprising precision to MI. Activation patching measures affinity: is this component engaged during the task? Ablation measures efficacy: does removing it change the output? But even ablation conflates efficacy with the system's compensatory capacity—a load-bearing head can show small ablation effects if backup mechanisms compensate. The observed effect magnitude is a joint property of the component's role and the network's reserve, and a single ablation cannot separate them. The practical implication is the *therapeutic window*: sweeping ablation strength from 0 to 1 reveals whether there is a range where on-task effects grow before off-target degradation begins. A circuit with no therapeutic window cannot be cleanly separated from the network's general processing.

Table 10: Pharmacology foundations.

Concept	**Source**	**Criterion**	**MI analog**
Dose-response curve	Hill 1910	E5	Partial ablation → partial effect
Affinity vs efficacy	Stephenson 1956	I1 vs I2	Participation ≠ causation
System reserve	Black & Leff 1983	I2	Backup circuits compensate after ablation
Functional selectivity	Kenakin 2004	E1	Ablation method determines the finding
Phase I/II/III trials	—	Verdict tiers	Calibrate → establish causality → generalize

5.5 Causal Inference

The internal validity criteria draw on causal inference methodology (Pearl, 2009; Spirtes et al., 2000). The necessity/sufficiency distinction maps to Woodward's interventionist causation (Woodward, 2003): necessity (I1) corresponds to "would the behavior persist if the component were absent?" and sufficiency (I2) to "would the component alone produce the behavior?" Pearl and Bareinboim's transportability theory (Pearl and Bareinboim, 2011) provides the formal conditions under which causal findings transfer across settings, grounding E4 (cross-model generalization). Rubin's potential outcomes framework grounds E5 (graded response) by formalizing heterogeneous treatment effects across input subpopulations—not all prompts may respond to ablation equally.

Table 11: Causal inference foundations.

Concept	**Source**	**Criterion**	**MI analog**
do-calculus / SCM	Pearl 2009	I1, I2	Formal language for ablation claims
Interventionism	Woodward 2003	I1	"Would behavior persist if component absent?"
Transportability	Pearl & Bareinboim 2011	E4	Conditions for circuit generalization across models
Causal discovery	Spirtes et al. 2000	—	Learning circuit structure from data
Potential outcomes	Rubin 1974	E5	Heterogeneous effects across input subpopulations

6 Case Studies

We apply the framework to 13 published MI results, evaluating each through all five validity lenses. The case studies are not intended to rank papers—they demonstrate what the framework looks like in practice and show that the tier system discriminates between well-validated and poorly-validated claims. Quantitative per-paper scoring—including the Claim Validity Score, tier-agreement confusion matrix, and per-criterion heatmaps—is the subject of a companion paper; here we keep the analysis qualitative and trace each verdict to specific missing or present evidence.

6.1 Verdict Assignments

Table 12 presents the verdict assignments. The key finding is that the tier system produces meaningful discrimination: the 13 claims span the full range from Proposed to Triangulated, with verdict differences traceable to specific missing evidence rather than arbitrary scoring.

Table 12: Verdict assignments for 13 published circuits, sorted by evidential status. Verdicts are based on published evidence as of evaluation date.

Verdict	**Circuit**	**Key evidence gap**
Validated	Grokking	Mathematically complete algorithm specification; every weight matrix entry predicted by closed-form Fourier formula
Triangulated	Induction Heads	Simple mechanism, broad replication, thick nomological network
Mechanistically Supported	Greater-Than	Strong structural plausibility; limited prompt generalization
Mechanistically Supported	Copy Suppression	Unusually clean specificity; sufficiency partially established
Mechanistically Supported	IOI Circuit	Method-conditional faithfulness (87% mean ablation, <50% other methods); specificity untested
Mechanistically Supported	Superposition	Toy-model-only evidence; criteria semi-confirmed rather than confirmed
Causally Suggestive	Successor Heads	Cross-domain generalization as convergent evidence; ablation method-conditional
Causally Suggestive	Othello World Model	"World model" label exceeds evidence ("linearly decodable"); interpretive inflation
Causally Suggestive	Docstring Circuit	Label ambiguity: "variable binding" vs. simpler "positional copying"
Causally Suggestive	Knowledge Neurons	Strong intervention but weak specificity; editing corrupts related facts
Causally Suggestive	SAE Features	Thin nomological network; features may be dictionary properties, not model properties
Proposed	Probing Classifiers	Decodability without intervention = no internal validity
Disconfirmed	Gender Bias Circuits	Construct incoherence: bias and knowledge share the same heads; discriminant validity (C4) fails—the construct itself is not well-posed

6.2 Illustrative Examples

We describe three case studies in detail to illustrate how the framework produces its verdicts.

Induction Heads (Triangulated). Olsson et al. (2022) identify a two-head composition mechanism for in-context copying: previous-token heads attend to the token before a repeated sequence, and induction heads use this signal to predict the next token. The claim reaches Triangulated because it satisfies criteria across multiple independent evidence families. The mechanism has been replicated across model scales and architectures (E4), confirmed through both activation-level and weight-level analysis (C3 convergent validity from different families), and produces clean dose-response effects (E5). The nomological network is thick: the mechanism connects to in-context learning theory, training dynamics, and cross-model structural predictions. The relative simplicity of the mechanism (two functional roles, clear compositional structure) makes each criterion easier to satisfy.

IOI Circuit (Mechanistically Supported). Wang et al. (2023) identify 26 attention heads forming a six-role circuit for indirect object identification. The circuit demonstrates strong necessity (I1) and sufficiency (I2): running the model with everything outside the circuit mean-ablated recovers 87% of the full model's logit difference. However, Miller et al. (2024) show that this 87% figure is specific to mean ablation; under other ablation methods, faithfulness drops below 50%. This method-conditional result partially satisfies E1 (intervention reach) but reveals that the causal claim is weaker than the headline number suggests. Specificity (I3) is untested: the authors do not report whether ablating the IOI circuit degrades unrelated tasks. A formal double-dissociation has not been conducted. The circuit reaches Mechanistically Supported rather than Triangulated because the evidence comes primarily from one methodological family (ablation-based) and key criteria (I3, I4, I5, E4) remain unaddressed.

Probing Classifiers (Proposed). Linear probing (Belinkov, 2022) trains a classifier on activations to test whether a concept is linearly decodable. The core limitation is fundamental: *a measurement without an intervention is a measurement without internal validity.* Probe success establishes that information is decodable from the activation space, but not that the model uses that information during inference. High probe accuracy is consistent with genuine encoding, incidental linear separability in high-dimensional space, and confound encoding (a correlated feature rather than the target). Without causal follow-up (DAS, intervention along the probe direction), the claim cannot advance beyond Proposed. Probing *with* causal follow-up can reach Causally Suggestive; with additional control tasks and cross-method convergence, it can reach Mechanistically Supported. The framework does not dismiss probing—it precisely characterizes what probing alone does and does not establish.

6.3 Recurring Validity Gaps

Several patterns emerge from evaluating these claims side by side.

Sufficiency gap. Most circuits demonstrate necessity (I1) but not sufficiency (I2). The I1→I2 transition—from "removing this breaks the behavior" to "this alone produces the behavior"—is the most common barrier between Causally Suggestive and Mechanistically Supported.

Method-conditional results. The IOI circuit's headline numbers are specific to mean ablation. Ablation type is part of the causal claim, not an implementation detail. The framework captures this through E1 (intervention reach), which requires agreement across ablation methods.

Toy-model ceiling. Grokking reaches Triangulated because the evidence is mathematically complete within its toy scope, while Superposition reaches Mechanistically Supported because its criteria are semi-confirmed rather than confirmed. Both illustrate how the framework handles scope: a claim validated within a declared scope is a legitimate result, not a failure (V5). The gap between toy-model proof-of-concept and real-model confirmation remains the field's central challenge.

Interpretive inflation. Labels like "world model," "knowledge neuron," and "deception feature" carry theoretical implications beyond what the evidence supports. The framework systematically identifies where labels exceed evidence through V4 (anthropomorphism check) and V5 (scope declaration).

Construct incoherence. Gender Bias Circuits are the only claim in our evaluation that reaches Disconfirmed—not because evidence is lacking but because the construct itself fails discriminant validity (C4). "Gender bias" and "legitimate gender knowledge" are implemented by the same attention heads; you cannot ablate one without ablating the other. No amount of additional evidence can fix a construct that is not separable from its neighbor. The framework surfaces this as a C4 failure, and the appropriate response is not more experiments but reformulation of the question.

7 Discussion

The case studies reveal that the dominant validity gaps in published MI research are systematic, not idiosyncratic. Sufficiency is rarely established. Specificity is rarely tested. Faithfulness results are method-conditional. Labels routinely exceed what the evidence supports. These are structural features of a field that evaluates methods and artifacts without a common protocol for evaluating the claims built on them.

The framework addresses this by providing a structured validity profile for each claim—making gaps visible and the path to strengthening them concrete. Organizations deploying MI-based monitors (deception probes, steering vectors, circuit-level classifiers) can evaluate the underlying features through the same criteria: Is the construct well-defined? Is the measurement reliable? Is the evidence causal? Does the finding generalize? A feature at Proposed should not be deployed as a monitor; a feature at Mechanistically Supported or above has the minimum evidential basis for cautious deployment with ongoing validation.

Several directions for future work follow naturally. Extending the case studies to larger models may reveal that criteria such as E4 (cross-model generalization) are harder to satisfy than the GPT-2 Small evaluations suggest. Automating the verdict assignment process—currently requiring human judgment in aggregating criterion-level results—is the subject of a companion paper. And developing standardized prompt suites for specificity testing (I3) and double dissociation (I4) would address the two most common gaps identified across the 13 case studies.

8 Limitations

This work is a theoretical contribution. The case study verdicts are based on published evidence, not new experiments; running the framework's full metric suite with actual model interventions is future work that requires GPU compute. The current evaluations are applied to GPT-2 Small, and extension to larger models may reveal that some criteria are harder to satisfy than the small-model case studies suggest.

While the 27 criteria are operationally defined, the aggregation from criterion-level results to verdict tiers involves judgment. Two evaluators could assign different tiers to the same circuit if they weight partial evidence differently. The framework mitigates this by requiring criterion-level transparency: the structured validity profile is the primary output, and the tier is a summary.

9 Conclusion

The MechVal framework provides a protocol for evaluating mechanistic interpretability claims. Its 27 criteria, organized across five validity types in dependency order, give each claim a structured validity profile that makes gaps visible and the path to strengthening them concrete. The framework does not rank papers or methods. It characterizes the evidential status of claims—and in doing so, it distinguishes what has been established from what has merely been asserted.

The framework is open-source and method-agnostic. It applies the same criteria to circuits, SAE features, probing classifiers, steering vectors, transcoders, and crosscoders—because the questions "Is the construct well-defined?" and "Is the evidence causal?" do not depend on how the evidence was produced.

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A Appendix

This appendix provides supporting material: an example circuit specification (Appendix A.1), the evidence card reporting format (Appendix A.2), and the full metric inventory grouped by evidence family (Appendix A.3).

A.1 Example Circuit Specification: IOI

We begin with a worked example to ground the abstract machinery: the IOI circuit, the most widely studied circuit in the field. Encoding it as a formal specification shows how a mechanistic claim becomes a set of concrete, falsifiable predictions that the verification pipeline can test automatically.

The IOI circuit specification defines a 6-step computational DAG:

1. **Duplicate Token Detection** (DTH: heads 0.1, 3.0) — detects that a name appears twice.
2. **Previous Token Tracking** (PTH: heads 2.2, 4.11) — attends to the token before a name.
3. **Induction** (IND: heads 5.5, 6.9) — uses DTH and PTH signals to identify the repeated name.
4. **S-Inhibition** (S-Inh: heads 7.3, 7.9, 8.6, 8.10) — suppresses the repeated subject name.
5. **Name Mover** (NM: heads 9.9, 9.6, 10.0) — copies the non-repeated (indirect object) name to the output.
6. **Negative Name Mover** (NegNM: heads 10.7, 11.10) — provides opposing signal.

Positive predictions (7).

- Ablating DTH reduces output (≥ 0.2 decrease in logit difference).
- Ablating induction heads substantially reduces logit difference (≥ 0.5).
- Ablating S-Inh kills output (≥ 0.8 decrease).
- Ablating S-Inh reduces name mover activation (≥ 0.2).
- Ablating NegNM increases logit difference (removes opposing signal).
- Ablating DTH reduces induction head activation (tests DTH$\rightarrow$IND edge).
- Ablating PTH reduces induction head activation (tests PTH$\rightarrow$IND edge).

Negative controls (5).

- Ablating NM does not affect upstream S-Inh.
- Ablating NegNM does not affect upstream S-Inh.
- Ablating PTH alone does not destroy circuit output (< 0.3 decrease).
- Ablating S-Inh does not affect upstream DTH.
- Ablating NM does not affect NegNM (parallel roles, not connected).

This specification encodes the directed causal structure of the proposed mechanism and provides concrete, falsifiable predictions that can be tested automatically by the verification pipeline.

A.2 Evidence Cards

An evidence card is the minimal reporting unit for a claim-level validity judgment. It records which measurement bears on which criterion, whether the relevant Family F calibration checks were performed, and what verdict implication follows. This prevents verdicts from becoming aggregate impressions: each assignment must be traceable to specific present or missing evidence.

As an example, we show the IOI circuit, using supplementary information from follow-up studies. (Miller et al., 2024) find the faithfulness numbers to be metho-conditional, and (Méloux et al., 2025) find alternative head sets achieving comparable faithfulness.

Table 13: Evidence card template: the atomic reporting unit linking measurements to criteria and verdicts.

Field	**Contents**
Claim	The exact mechanistic claim being evaluated
Description Mode	The strongest description mode implicated by the claim
Evidence Family	The family of the primary evidence: causal, structural, information-theoretic, behavioral, representational, or measurement
Primary Metric	The concrete runnable test producing the measurement
Family F Checks	Calibration checks on the primary metric: reliability, baseline separation, stability, sensitivity, uncertainty, or invariance
Criterion Tested	The criterion the evidence bears on, such as I1 necessity, I2 sufficiency, E1 intervention reach, or M2 baseline separation
Result	The reported numerical result, qualitative finding, or structured observation
Status	Pass, partial pass, fail, or untested
Failure Mode Addressed	The validity failure this card helps detect, such as no baseline control, method-conditional evidence, off-target effects, or scope creep
Verdict Implication	What verdict tier this evidence can support, and what higher tier it fails to support without additional evidence

Table 14: Evidence card for the IOI circuit faithfulness result.

Field	**IOI example**
Claim	The IOI circuit uses name-mover heads to copy the indirect object to the output position
Description Mode	Implementational-functional: the claim identifies components and assigns them functional roles
Evidence Family	Behavioral and causal evidence
Primary Metric	Faithfulness / logit-difference recovery under circuit ablation
Family F Checks	Intervention robustness: the headline faithfulness value is method-conditional
Criterion Tested	I1 necessity, I2 sufficiency, and E1 intervention reach
Result	The circuit recovers 87% of the full-model logit difference under mean ablation, but substantially less under other ablation methods
Status	I1 pass; I2 partial/pass depending on intervention definition; E1 partial because intervention variants disagree
Failure Mode Addressed	Cherry-picking metrics and method-conditional evidence
Verdict Implication	Supports Mechanistically Supported, but blocks Triangulated until specificity, double dissociation, cross-model generalization, and stronger measurement calibration are addressed

A.3 Metrics Inventory

The full inventory of metrics is listed below, grouped by evidence family: causal, structural, information-theoretic, behavioral, representational, and measurement.

A. Causal

Table 15: Causal metrics: the effect of interventions on the model's computation.

Metric	**Description**
A01 SCM / Pearl	Structural causal models and do-calculus as the formal language for circuit claims
A02 Counterfactual DAS	Distributed Alignment Search and IIA for testing causal abstraction hypotheses
A03 Rubin CATE	Conditional average treatment effects across input subpopulations
A04 Woodward	Interventionist theory of causation applied to circuit ablation
A05 MDC / Glennan	New mechanical philosophy — components organized to produce phenomena
A06 Mediation	Natural direct and indirect effects decomposition for circuit paths
A07 Granger / TE	Observational causal discovery via conditional MI and temporal precedence
A08 PID	Decomposing MI into redundant, unique, and synergistic contributions
A09 MDL / SLT	Minimum description length and singular learning theory for circuit complexity
A10 Regularity / INUS	INUS conditions — insufficient but necessary parts of sufficient sets
A11 Actual Cause	Halpern-Pearl actual causation on specific inputs, not just potential causes
A12 Transportability	Formal conditions for circuit generalization across models and distributions
A13 Causal Discovery	NOTEARS / PC algorithms for learning DAGs over circuit components

B. Structural

Table 16: Structural metrics: weight-space properties requiring no forward pass.

Metric	**Description**
B01 SVD / Spectral	Singular value decomposition to identify dominant computational directions
B02 Effective Rank	Entropy-based dimensionality as a scalar summary of spectral concentration
B03 OV / QK Decomp.	Decomposing attention heads into OV (what to write) and QK (where to attend)
B04 Weight Alignment	Cosine similarity between principal weight directions across heads
B05 Norm Trajectory	Spectral norm ratios tracking signal amplification through components
B06 Template Distance	Graph-edit and metric distances between circuits discovered for different tasks
B07 Polysemanticity	Measuring whether components encode multiple unrelated features in superposition
B08 ICA / NMF	Independent component analysis for decomposing weights into interpretable parts
B09 Weight Classifier	Training classifiers on weight matrices to predict circuit membership

C. Information-theoretic

Table 17: Information-theoretic metrics: information flow and dependencies.

Metric	**Description**
C01 Mutual Info.	Total shared information between circuit components and task performance
C02 Conditional MI	Information shared after conditioning on other parts of the circuit
C03 Transfer Entropy	Directed information flow between components across layers
C04 PID	Decomposing shared information into unique, redundant, and synergistic atoms
C05 Info. Bottleneck	How efficiently circuits compress input while preserving task-relevant signal
C06 O-Information	Whether a group of components interacts redundantly or synergistically
C07 Granger Causality	Whether one component's past activations improve prediction of another's future
C08 OCSE	Estimating causal influence between components using only observational data
C09 NOTEARS	Learning directed acyclic graph structure from observational activation data

D. Behavioral

Table 18: Behavioral metrics: input-output behavior under manipulation.

Metric	Description
D01 Faithfulness	Whether an identified circuit faithfully reproduces the full model's behavior
D02 Logit Diff	How much of the model's logit difference the circuit recovers
D03 KL Divergence	Information-theoretic distance between circuit and full model distributions
D04 CE Delta	Change in cross-entropy loss when the circuit is ablated
D05 Top-K Accuracy	Whether the circuit preserves the model's top-K predicted tokens
D06 Cross-Task	Whether a circuit discovered on one task transfers to a different task
D07 Cross-Scale	Whether circuit structure replicates in larger or smaller models
D08 Prompt Paraphrase	Circuit consistency across semantically equivalent prompt templates
D09 Generalization Gap	Sensitivity of circuit discovery to hyperparameters and methodological choices

E. Representational

Table 19: Representational metrics: geometry and content of internal representations.

Metric	Description
E01 DAS-IIA	Whether a learned linear subspace causally encodes a target variable
E02 Linear Probe	Whether a target variable is linearly decodable from intermediate representations
E03 RSA	Comparing representation geometry by correlating pairwise distance matrices
E04 CKA	Kernel alignment for cross-layer and cross-model representation comparison
E05 Subspace Align.	Cosine alignment between SVD-derived principal directions of weight matrices
E06 PCA Dim.	Effective dimensionality of circuit subspaces via activation covariance spectrum
E07 Intrinsic Dim.	True manifold dimensionality, connecting to geometric complexity
E08 Participation Ratio	How many dimensions are effectively active in a representation
E09 Persistent Homology	Topological data analysis detecting loops and voids in activation manifolds
E10 Cross-Task Overlap	Representational structure shared between tasks via IIA transfer

F. Measurement

Table 20: Measurement metrics: calibration checks for the trustworthiness of other metrics.

Metric	**Description**
F01 Test–Retest	Whether rerunning the same metric on the same model, task, and intervention gives the same answer
F02 Bootstrap Stability	Whether the metric remains stable under bootstrap resampling of prompts, examples, or activation samples
F03 Seed Variance	How much the metric changes across random seeds for probes, SAEs, optimization, or sampling-based estimators
F04 Checkpoint Variance	Whether the metric is stable across nearby checkpoints or depends on one training snapshot
F05 Prompt Variance	How much the metric changes across prompt templates, paraphrases, lexical choices, or dataset slices
F06 Baseline Separation	Whether the score is distinguishable from random, untrained, shuffled-label, or permuted-circuit baselines
F07 Negative Controls	Whether the metric stays low where the claimed effect should be absent
F08 Positive Controls	Whether the metric detects known-true or synthetic effects of the relevant size
F09 Intervention Robustness	Whether the conclusion survives reasonable intervention variants such as mean ablation, zero ablation, resample ablation, patching, or causal scrubbing
F10 Hyperparam. Sensitivity	Whether the metric changes under reasonable choices of sparsity, probe regularization, thresholds, localization cutoffs, or discovery hyperparameters
F11 Estimator Uncertainty	Confidence intervals, standard errors, permutation tests, or posterior intervals for the reported metric value
F12 Calibration Curve	Whether larger reported scores correspond to larger empirical effects or higher probability of success
F13 Cross-Metric Convergence	Whether independent metrics intended to test the same criterion agree despite different failure modes
F14 Measurement Invariance	Whether the metric behaves consistently across model sizes, tasks, prompt distributions, and intervention contexts